LE NGUYEN DUY LUAN

Android & AI Engineer

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Portfolio | GitHub | LinkedIn

Professional Summary

Mobile Developer specializing in Kotlin, Jetpack Compose, and on-device AI applications. Experienced in designing offline-first Android apps with modern architecture (Clean Architecture, MVVM, Room, CameraX) and deploying TensorFlow Lite models for real-time edge inference. Passionate about building performant applications that combine excellent user experience with practical AI capabilities.

Education

Nha Trang University 2022 – 2026
Bachelor of Information Technology *Nha Trang, Vietnam*

- **GPA**: 3.25/4.0

Technical Skills

Programming Languages: Kotlin, Java, Python, SQL, JavaScript
Android: Jetpack Compose, CameraX, Room, WorkManager, Navigation, Paging3, DataStore
Architecture: Clean Architecture, MVVM, Repository Pattern, Dependency Injection (Hilt), StateFlow, Coroutines
AI & CV: TensorFlow Lite, MediaPipe, ML Kit, OpenCV, YOLO (Classification)
Backend & Cloud: Firebase, Supabase, REST API, Retrofit, OkHttp
Tools: Git, Android Studio, Gradle, ADB, KSP, VS Code
Languages: Vietnamese (Native) – English (Intermediate, comfortable reading technical docs)

Projects

VocabLensAI — Native Android English Learning Platform | *Personal Project*

- **Tech Stack**: Kotlin, Jetpack Compose, Coroutines/Flow, Room DB, CameraX, TensorFlow Lite, Gemini API, Firebase, Supabase.
- Designed and developed a production-scale offline-first Android application using Kotlin, Jetpack Compose, MVVM, and Clean Architecture, supporting AI-powered vocabulary learning with real-time object recognition.
- Implemented on-device object detection using CameraX and TensorFlow Lite to recognize real-world objects in real time, integrating Gemini AI to generate contextual vocabulary stories.
- Addressed the "forgetting curve" problem in language learning by designing a custom SM-2 Spaced Repetition engine that automatically schedules reviews at optimal intervals, backed by a hybrid Room DB/Firestore sync system to ensure reliable offline data sync.
- Built a real-time multiplayer PvP Arena using Firebase Firestore with an ELO rating calculator, alongside a chess-style game review UI for match accuracy analysis and AI feedback.

Lung X-ray Trustworthy Classifier (Android + Research) | *Graduation Project, NTU* 9.1/10

- **Tech Stack**: Python, TFLite, U-Net (ResNet50), YOLOv12m-cls, Kotlin, Jetpack Compose, Groq API.
- Deployed a 2-stage AI pipeline (U-Net segmentation → YOLOv12m-cls image classification) to a native Android app via TFLite, enabling fully offline on-device inference across 4 pathology classes (COVID-19, Viral Pneumonia, Lung Opacity, Normal).
- Improved model reliability by engineering median-fill masking to remove non-lung visual artifacts before classification, increasing explainability scores and localization accuracy across 300 annotated cases.
- Built an OOD safety gate (lung-area ratio thresholding, 8–72% valid range) to auto-reject invalid inputs, and integrated dual-mode XAI (Occlusion Sensitivity + Grad-CAM++).
- Added a context-aware LLM assistant (Groq/LLaMA 3.1, Fleischner-guideline grounded) and native PDF report export, benchmarked against 5 backbone architectures.

AI Fitness Coach | *Personal Project*

- **Tech Stack**: Kotlin, Jetpack Compose, CameraX, MediaPipe/MLKit (Pose Estimation), Room DB, Text-To-Speech (TTS).
- Developed an offline-first Android app (MVVM) providing a comprehensive ecosystem with an 800+ exercise library and an AI-driven personalized workout recommendation engine.
- Addressed the injury risk of unsupervised home workouts by implementing camera-based pose estimation (MediaPipe) to extract 33 keypoints and calculate 3D vector angles in real-time (60 FPS, ¡20ms latency).
- Built a custom rule-based feedback engine with Text-To-Speech integration (audio alerts) and automated rep counting, caching progress 100% offline via Room DB.